DATABASES

# Redis Visual Guide

Data structures, persistence, pub/sub -- the in-memory Swiss army knife.

**ANIMATED WALKTHROUGH**

Step through pages to play the animation

STEP 1/6

# Data Structures

Not just strings -- a toolkit of types.

### Strings

SET / GET -- cache values, counters (INCR)

### Hashes

HSET user:1 name Ada -- objects, field-level ops

### Lists / Sets

LPUSH / SADD -- queues, feeds, tags, dedupe

### Sorted Sets

ZADD leaderboard 1500 ada -- rankings, rate limits

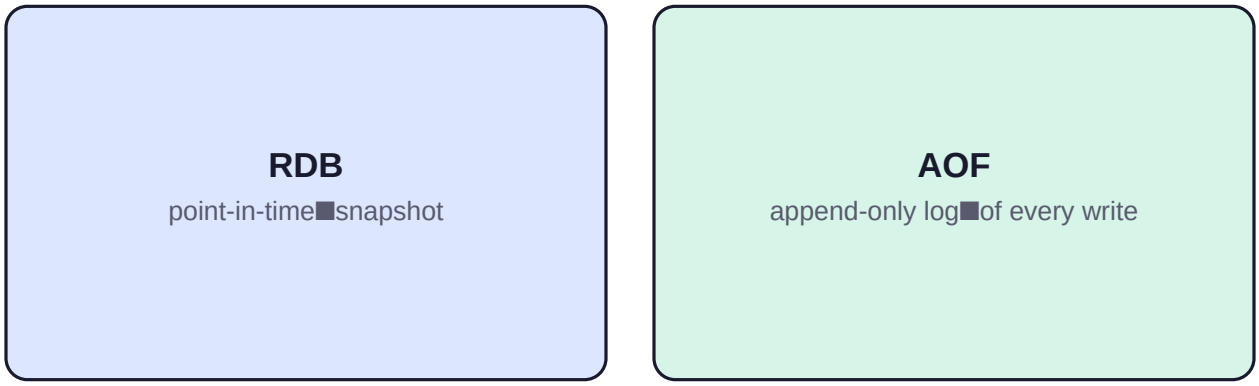
### Streams

XADD events * -- durable log, consumer groups

STEP 2/6

# Persistence: RDB vs AOF

Memory is fast; disks remember.




- RDB: compact, fast restarts -- but lose minutes of data on crash.
- AOF (fsync everysec): lose ~1s max -- bigger files, slower rewrites.
- Production default: BOTH. RDB for fast boot, AOF for durability.

STEP 3/6

# Pub/Sub & Streams

Fire-and-forget messaging vs durable logs.

Pub/Sub: no persistence -- offline subscribers MISS messages.

**Streams (XADD / XREADGROUP)**

durable append-only log + consumer groups -- Kafka-lite inside Redis

Need guaranteed delivery? Streams. Need live fan-out? Pub/Sub.

STEP 4/6

# Caching Patterns

Cache-aside: the workhorse pattern.

- Read: check cache -> miss -> read DB -> SET with TTL.
- Write: update DB, then DELETE the cache key (invalidate, don't update).
- Thundering herd: one miss x 10k requests -- use locks or request coalescing.

STEP 5/6

# Operating Redis

Eviction, expiry, and clustering.

### Eviction policies

allkeys-lru / volatile-ttl / noeviction -- pick per use case

### TTL everything

keys without expiry are memory leaks with extra steps

### Cluster hash slots

16384 slots sharded across nodes -- multi-key ops must share a slot

Monitor used_memory + evicted_keys: evictions climbing means the working set outgrew RAM.

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# Key Takeaways

## Redis -- Remember

- Pick the right structure: strings, hashes, sets, sorted sets, streams -- each is a tool.
- Persist with RDB + AOF together: fast restarts plus ~1s durability.
- Streams give durable consumer-group messaging; pub/sub is ephemeral fan-out.
- Cache-aside: read-through on miss, invalidate (not update) on write, TTL everything.
- Watch memory: eviction policy + TTLs decide whether Redis stays fast or starts shedding.

**Keep exploring**

Pair this guide with the matching video walkthroughs and hands-on labs at interstitiumlabs.dev